# CLASS 6 — MATHS · FORMULA SHEET

Chapter 10: **The Other Side of Zero** | Quick revision, one page

## Integers and the number line

- Integers: $\ldots, -3, -2, -1, 0, 1, 2, 3, \ldots$ Zero is neither positive nor negative.
- On the number line, numbers increase to the right. Every positive integer $> 0 >$ every negative integer.
- Successor $= n + 1$; predecessor $= n - 1$. Opposite of $n$ is $-n$, and $n + (-n) = 0$.

## Comparing

- Of two negative integers, the one with the smaller absolute value is greater: $-3 > -7$.
- Ascending order: smallest to greatest (left to right on the number line).

## Addition

- Same signs: add the values and keep the sign: $(-4) + (-3) = -7$.
- Different signs: subtract the smaller value from the larger and keep the sign of the larger: $(-4) + 6 = 2$, $(-9) + 5 = -4$.

## Subtraction

- $a - b = a + (-b)$: subtracting an integer is adding its opposite.
- $a - (-b) = a + b$, e.g. $3 - (-4) = 7$. Change (rise or fall) $=$ final $-$ initial.

## Applications

- Temperature below zero, depths below sea level and floors below ground are written with a negative sign.
- Profit $= +$, loss $= -$; overall result $=$ sum of all the amounts.

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